Day 4: Other Protocols in Action – FTP and SNMP

Task 1: Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Objective:
To install FileZilla Server on a Windows machine and create a user account with access to a shared folder.

Procedure:

1. Download FileZilla Server from its official website.
2. Run the installer and complete the installation.
3. Open the FileZilla Server Administration Interface.
4. Create a new FTP user account with a username and password.
5. Create a folder on the computer that will be used for file sharing.
6. Add this folder as a shared directory for the FTP user.
7. Assign appropriate permissions, such as reading and writing files.

Expected Result:
FileZilla Server is installed and configured with a user account and a shared folder for FTP file access.

Task 2: Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.

Objective:
To connect to the FTP server using FileZilla Client and upload an image file.

Procedure:

1. Download and install FileZilla Client.
2. Open FileZilla Client.
3. Enter the FTP server's IP address, username, password, and port number.
4. Click the Quickconnect button to establish a connection.
5. Locate an image file on the local computer.
6. Navigate to the shared folder on the server.
7. Drag and drop the image file into the server's directory.

Expected Result:
The FileZilla Client successfully connects to the FTP server, and the selected image file is uploaded to the shared folder.

Task 3: Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

Objective:
To download the uploaded image from the FTP server and verify its file size.

Procedure:

1. Open FileZilla Client and connect to the FTP server.
2. Navigate to the shared folder containing the uploaded image.
3. Create a different folder on the local computer for downloading the file.
4. Select the uploaded image from the server.
5. Download it into the newly created local folder.
6. Compare the original image file size with the downloaded image file size.

Expected Result:
The image file is successfully downloaded, and its file size matches the original file size, confirming that the transfer was successful.

Task 4: Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

Objective:
To identify three real-world scenarios where SNMP is used in network management.

What is SNMP?

SNMP stands for Simple Network Management Protocol. It is used by network administrators to monitor and manage network devices such as routers, switches, servers, and access points.

Real-World Applications:

1. Internet Service Providers (ISPs)

- ISPs use SNMP to monitor routers and network devices.
- It helps administrators check bandwidth usage and network traffic.
- It can help identify device failures and connectivity problems.

2. Campus Wi-Fi Networks

- Educational institutions can use SNMP to monitor wireless access points and network switches.
- It helps track network performance and device availability.
- Administrators can identify overloaded access points and connectivity issues.

3. Large Websites and Data Centers

- Data center administrators can use SNMP to monitor network switches, routers, and servers.
- It helps monitor device health, network traffic, and interface status.
- It helps administrators detect network problems and maintain service availability.

Expected Result:
SNMP helps network administrators monitor device performance, bandwidth usage, and network availability in different environments.

Task 5: Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

Objective:
To compare FTP and HTTP based on their features and usage.

Comparison Point	**FTP**	**HTTP**
Full Form	File Transfer Protocol	Hypertext Transfer Protocol
Port Number	21 (control connection); data uses port 20 in active mode or negotiated ports in passive mode	80 (default HTTP)
Security	Standard FTP transfers credentials and data without encryption	HTTP does not encrypt data; HTTPS provides encryption
Data Transfer	Transfers files between client and server	Transfers web pages, images, videos, and other web resources
Use Cases	Uploading and downloading files	Accessing websites and web applications
Authentication	Supports username and password authentication	Basic HTTP does not require login; web applications can use different authentication methods
Main Purpose	File sharing and transfer	Communication between web browsers and web servers

Conclusion:
FTP is mainly designed for transferring files between computers, whereas HTTP is designed for transferring web content between browsers and web servers. Both protocols play important roles in network communication.

management were explored, and FTP was compared with HTTP based on their features and functionality.